

NCIE-015 Skeletal v0.1 → Full v1.0

Expansion Verification Report

DevSecOps, Infrastructure & Deployment Specification — IN DEVELOPMENT / FOR HUMAN REVIEW OF FULL EXPANSION

1. Title Reconciliation Verification (Directive Item 3) — SOURCE-VERIFIED

Authoritative register used: NCIE Documentation Suite Master Controlled Register v2.0 (FOR APPROVAL, dated 02 September 2026), located on the controlled document store. NCIE-015 entry located: Section 3, 'Reconciled Master Documentation Register' table, row NCIE-015. Authoritative formal title (as recorded in the register): 'DevSecOps, Infrastructure & Deployment Specification'. Skeletal source-supported working title prior to this verification: 'NCIE-015 — DevSecOps, Infrastructure & Operations Specification'. Comparison result: SEMANTIC MATCH — DOCUMENT-CONTROL WORDING DIFFERS (the register's Primary Purpose text — 'Environments, infrastructure, CI/CD, observability, resilience and deployment' — matches this document's actual 36-chapter operational scope exactly; the only difference is the final word, 'Deployment' in the register versus 'Operations' in the skeleton's working title; no material discrepancy in scope, responsibility or substance was found). Correction applied: the document-control title was set to 'NCIE-015 — DevSecOps, Infrastructure & Deployment Specification' in the cover, §0.1, the Document Governance table, the Authority Chain, the running header, §36.6, core document properties and the deliverable filename, per directive item 3's wording-correction procedure (document-control only, no Human Review halt required). 36-chapter integrity: PASS (36/36). Salient Things to Note: PASS (36/36). Human Review Register reconciliation: PASS (36 total / 32 Blocking / 4 Non-Blocking / 0 duplicate / 0 orphan). Three-artifact consistency: PASS. This finding is SOURCE-VERIFIED — established directly from the text of the Master Controlled Register v2.0 document itself and the skeletal document's own Document Governance block, not inferred from naming convention or conversational context.

2. Scope Verification

NCIE-015 v1.0 is a controlled full expansion of the approved NCIE-015 Skeletal v0.1: every approved chapter purpose, Salient Things to Note, principal component, upstream dependency/authority and Human Review focus is elaborated into environment/promotion, IaC, source control, build/supply-chain, CI gates, release management, deployment strategies, runtime/Agent-sandbox/network/IAM/secrets operations, database/storage and integration operations, core-service/module/front-end delivery operations, observability/SLO/alerting/incident management, runbooks, change/capacity/backup/DR operations, vulnerability/security operations, operational audit/reconciliation, go-live readiness and DevSecOps testing, without inventing IaC/source-control/CI/CD/artifact/orchestration/observability/incident/secrets/infrastructure products or SLOs/RPO/RTO/alert thresholds/capacity limits/remediation deadlines/maintenance windows/retry counts/backup schedules/exercise cadence absent source authority.

3. 36-Chapter Integrity Verification

A programmatic scan of the final document confirms exactly 36 sequential 'NCIE-015 — Chapter N Expansion' labels, N = 1..36, each present exactly once, in the approved order, with no chapter added, removed, merged, split or renamed relative to the approved Skeletal v0.1 structure.

4. Salient Things to Note Hard-Gate Verification

Per directive item 78 — continuing the discipline established after the NCIE-009 Chapter-27 omission — every chapter must retain an explicit '<N>.0 Salient Things to Note' subsection. This build enforced the requirement structurally: the chapter-rendering helper (lib.py's add_chapter function) raises a build-time assertion error if a chapter's salient content is omitted, so the document cannot be assembled with a missing subsection. A programmatic scan of the final assembled document, rerun independently rather than inferred from the build script succeeding, confirms an explicit Salient Things to Note heading present for all 36 chapters (N = 1 through 36), with zero missing.

5. Skeleton Fidelity Verification

Each of the 36 chapters was authored directly from its approved skeletal Chapter Purpose, Salient Things to Note, Principal Components, Upstream Dependencies/Authority and Human Review Focus (see the accompanying Change Register for the per-chapter mapping). No approved chapter concept was omitted, and none was conceptually inverted.

6. No-Platform-Invention Audit (Directive Item 9)

A full-text scan for unsourced IaC, source-control, CI/CD, artifact, orchestration, observability, incident, secrets or infrastructure product selections found none — every platform decision (Chapters 4, 6, 10-11, 16, 20) is stated as a Human Review item (e.g. HR15-4-1, HR15-6-1, HR15-10-1, HR15-16-1, HR15-20-1) rather than a mandated product.

7. No-Numeric-Invention Audit (Directive Item 10)

A full-text scan for invented SLOs, RPO/RTO, alert thresholds, capacity limits, remediation deadlines, maintenance windows, retry counts, backup schedules or exercise cadence found none — Chapter 21 defers SLO targets (HR15-21-1), Chapter 22 defers alert thresholds (HR15-22-1), Chapter 26 defers capacity limits (HR15-26-1), Chapter 27 defers RPO/RTO (HR15-27-1) and Chapter 29 defers remediation targets (HR15-29-1) — every concrete number is a named Human Review item, never an invented default.

8. Release-Authority Audit (Directive Item 66)

A full-text scan for wording that lets build/pipeline/deployment/health state self-authorize production release or institutional acceptance found none — Chapter 8's Release-Control Matrix (Table 11) explicitly distinguishes build, CI-validation-passed, release candidate, Human/Governance authorization, deployed, health-verified and final-acceptance as seven non-substitutable states, and DEPLOYED ≠ APPROVED is restated at every chapter that touches release/deployment.

9. Emergency-Authority Audit (Directive Item 67)

Break-glass (Chapter 13), incident command (Chapter 23), runbooks (Chapter 24) and emergency changes (Chapter 25) were each verified to remain scoped, temporary and audited — §25.2 explicitly requires mandatory post-review and either ratification or reversion for every emergency change, and Table 30's Runbook Matrix fixes a stop/escalation condition for every scenario so automation cannot silently expand its own authority.

10. Current-State Recovery Audit (Directive Item 68)

Restore (Chapter 27), failover (Chapter 28) and rollback (Chapters 8-9) were each verified to require reconciliation of current identity, authorization, eligibility, policy, Evidence and approval state before being treated as authoritative — Table 28's Incident/Recovery Matrix (Chapter 23) makes this reconciliation requirement explicit for every incident/failover/restore state, consistent with NCIE-014's identical §23.2/§24 discipline.

11. Telemetry Audit (Directive Item 69)

A full-text scan confirms metrics, logs, traces and alerts are never represented as audit/provenance, root cause, a Finding, or business correctness — Chapter 20's Observability Matrix (Table 24) draws an explicit separation statement for every telemetry type, and Chapter 31 states plainly that telemetry is not audit/provenance.

12. Automation Audit (Directive Item 70)

Autoscaling (Chapter 26), auto-remediation (Chapter 32), pipeline automation (Chapters 6-9) and runbook automation (Chapter 24) were each verified to remain bounded by approved authority and exception state — AUTOSCALING ≠ CAPACITY ASSURANCE and RUNBOOK EXECUTION ≠ POLICY EXCEPTION AUTHORITY are both restated at their owning chapters, and §32.2 requires every auto-remediation repair to be proposed, approved and provenance-recorded.

13. Security Operations Audit (Directive Item 71)

A full-text scan confirms operator/infrastructure privilege is never represented as content authority (Chapter 13, 28) and that secrets/credentials are delivered by reference only, never embedded in code, IaC or logs (Chapter 14, 4) — Agent Context secret exposure is explicitly out of scope per Chapter 11's sandbox-boundary discipline.

14. Implementation-Claims Audit (Directive Item 74)

A full-text scan for language claiming pipelines, environments, runbooks, monitoring, DR, restore, security response or go-live controls are implemented or validated found none — every chapter's status line reads 'IN DEVELOPMENT — FOR HUMAN REVIEW' and Chapter 34's test-status framework (§34.2) marks every test class SPECIFIED only, per directive item 65's six-state distinction (TEST SPECIFIED, IMPLEMENTATION EVIDENCE AVAILABLE, TEST EXECUTED, TEST PASSED, GAME-DAY/RECOVERY VALIDATED, HUMAN/PRODUCTION ACCEPTANCE).

15. Adversarial Review (Directive Item 75)

The document was reviewed from DevSecOps, SRE/operations, platform-engineering, security-operations, incident/recovery, persistence/integration and governance perspectives. Corrections applied during review: ensured Chapter 23's Incident/Recovery Matrix explicitly covers post-incident review as a distinct state (not only detection through recovery), closing a gap a purely technical read of the skeleton's incident lifecycle would have missed; ensured Chapter 8's Release-Control Matrix distinguishes 'deployed' from 'health-verified' as separate states, since a deployment that is merely running is not yet confirmed healthy. No defect exposing private chain-of-thought was found or is reported.

16. Skeleton Fidelity Audit

All 36 approved Chapter Purposes, Salient Things to Note, Principal Components, dependencies and Human Review foci were verified represented and not inverted, per the mapping in the accompanying Change Register.

17. 36-Chapter Integrity Audit (Directive Item 77)

Programmatically re-verified: exactly 36 sequential chapters, no missing/duplicate/added chapter and no unintended renaming.

18. Salient Things to Note Audit (Directive Item 78)

Programmatically re-verified: an explicit Salient Things to Note subsection is present in all 36 chapters.

19. Human Review Register Mechanical Reconciliation

Chapter 35's register was recomputed programmatically by scanning the Human Review Focus bullet text already rendered in Chapters 1-34, plus HR15-35-1 (Chapter 35's own item) and HR15-36-1 (Chapter 36's own item) — both defined once and included in the register-construction list from the start of the build, a design choice made specifically to avoid repeating the register-reconciliation defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Result: 36 total items, 32 Blocking / 4 Non-Blocking, 0 duplicate Review IDs, 0 orphan Human Review items. Classification breakdown: 17 Institutional Confirmation Required, 9 Security/Sovereignty Confirmation Required, 8 Engineering/Infrastructure Discovery Required, 1 Legal/Policy Confirmation Required, 1 Resolved by Upstream Baseline.

20. Upstream Human Review Deduplication (Directive Item 64)

Every NCIE-015 Human Review item was individually checked against the existing NCIE-008–014 Human Review registers. None was found to restate an identical governance, security, persistence or domain-business question — each concerns a distinct operations decision those upstream documents raise the need for but do not themselves resolve. No cross-reference deduplication was required for this edition.

21. Blocking Scope Discipline

Each Blocking item in the Chapter 35 register states precisely which platform selection, numeric threshold, on-call/runbook authority or ownership assignment it blocks, rather than implying the whole document or the whole approved upstream chain is blocked. Non-Blocking items (Chapters 5, 9, 19, and the resolved Chapter 36 item) are explicitly marked as not gating the corresponding chapter's operational activity in the interim.

22. Environment/Promotion, Release-Control, Observability, Incident/Recovery and Runbook Matrix Verification (Directive Items 57-58, 60-62)

Table 5 (Chapter 3) was verified to state a data/credential policy, isolation requirement, deployment source, promotion evidence and prohibited practice for all four environment classes. Table 11 (Chapter 8) was verified to distinguish seven release states, each with its own authority and an explicit 'never implies' caution. Table 24 (Chapter 20) was verified to separate five telemetry/record types with an explicit audit/provenance-separation statement for each. Table 28 (Chapter 23) was verified to map six incident/recovery states to command authority, containment, reconciliation, Evidence preservation and acceptance. Table 30 (Chapter 24) was verified to fix a trigger, prerequisite, authority boundary, stop/escalation condition and evidence/rollback path for five representative operational scenarios.

23. Operational Ownership Matrix Verification (Directive Item 59)

Table 2 (Chapter 1) was verified to map every platform domain introduced later in the document (environments, IaC, source control/CI/build, release/deployment, runtime/network/Agent sandbox, IAM/secrets, database/storage, integration/core-service/module/frontend, observability/SLO/alerting/incident, and the remaining procedural/capacity/backup/DR/vulnerability/SecOps/audit/reconciliation/readiness/testing domains) to a technical operator responsibility and an upstream governance authority, without inventing an organizational office.

24. Character-Encoding Integrity Check

Following the same build pipeline used for the whole NCIE series (UTF-8 source fragments executed via a Python `exec()` call with an explicit `encoding='utf-8'` argument, a fix first required during the NCIE-008 build), a full-document scan of the final NCIE-015 document confirmed zero mojibake occurrences across all paragraphs and tables.

25. Visual and Structural Verification

The complete document was opened in Microsoft Word, its Table of Contents field (a live Word field, not static text) was updated and confirmed to populate with correct chapter/section titles and real page numbers, repaginated to 61 pages, saved, and rendered to PDF. Independent text extraction from the rendered PDF confirms the cover occupies page 1 alone and Document Governance begins on page 2 (the manual page break), with the running header and footer correctly reading NCIE-015 branding and the reconciled title. No table clipping, overlapping text, or broken table was observed across the six consolidated matrices, the Chapter 35 Human Review Register tables, or the Chapter 36 traceability/handover tables. A targeted scan for the stale wording 'Infrastructure & Operations Specification' used as a formal title (rather than the intentionally preserved historical working-title quote) found zero occurrences.

26. Final Human Review Extraction (Authoritative)

Performed directly against the final specification file by independently scanning every paragraph and table cell for the pattern HR15-<chapter>-<n>: TOTAL IDS FOUND = 36. UNIQUE IDS = 36. DUPLICATES = 0. OPEN = 35. RESOLVED = 1 (HR15-36-1, Resolved by Upstream Baseline). BLOCKING = 32. NON-BLOCKING = 4. CLASSIFICATION TOTALS: Institutional Confirmation Required = 17; Security/Sovereignty Confirmation Required = 9; Engineering/Infrastructure Discovery Required = 8; Legal/Policy Confirmation Required = 1; Resolved by Upstream Baseline = 1. ORPHAN IDS = 0 — every one of the 36 unique IDs found anywhere in the document also appears as a row in the Chapter 35 Complete Register table, which itself contains exactly 36 rows with no duplicate Review ID.

27. Three-Artifact Consistency Check

The NCIE-015 v1.0 main document, this Verification Report and the accompanying Full-Expansion Change Register agree exactly on: Formal title 'NCIE-015 — DevSecOps, Infrastructure & Deployment Specification' (reconciled at production time per §1 above, SOURCE-VERIFIED against Master Controlled Register v2.0); Source Version v0.1 (Skeletal) → Target Version v1.0 (Full); Authority chain approved NCIE-001 v1.3 through approved NCIE-014 v1.0, plus the approved NCIE-015 Skeletal v0.1; Status IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION (not Approved); 36 chapters preserved in order (0 missing, 0 duplicate, 0 added, 0 unauthorized renaming); Salient Things to Note hard gate = 36/36 explicit subsections present; Human Review Register TOTAL = 36, BLOCKING = 32, NON-BLOCKING = 4, classification totals as in §26 above; orphan Human Review IDs = 0; duplicate Human Review IDs = 0; DevSecOps/operations test-class status = 10 tests specified, none executed, passed, or game-day/recovery-validated; DevSecOps/operations invariants = 18 NCIE-015-specific plus cross-referenced upstream invariants; downstream handover = NCIE-016 and subsequent controlled documents each receive a defined interface without technical preemption; title-reconciliation result = SEMANTIC MATCH / DOCUMENT-CONTROL WORDING DIFFERS, corrected. No mismatch was found between the three artifacts.

28. Final Readiness Disposition

NCIE-015 — DevSecOps, Infrastructure & Deployment Specification v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed platform selection, numeric threshold, on-call model, runbook authority and ownership assignment deferred in Chapters 1-34 remains a proposal pending resolution of the Chapter 35 Human Review Register (including the formal-title confirmation, HR15-1-1); every §34.1 test remains SPECIFIED, not executed, passed, or game-day/recovery-validated, pending NCIE-016. This document is NOT marked Approved.